SCHOLARSHIP REQUEST MANAGEMENT SYSTEM USING SERVICENOW

A PROJECT REPORT

Submitted by

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BONAFIDE CERTIFICATE

Certified that this project report titled **"SCHOLARSHIP REQUEST MANAGEMENT SYSTEM USING SERVICENOW"** is the bonafide work of **ANSHU KUMAR (UID: 24BCS10580)** who carried out the project work under our supervision. Certified further that to the best of my knowledge the work reported herein does not form part of any other project report or dissertation on the basis of which a degree or award was conferred on an earlier occasion on this or any other candidate.

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I, **ANSHU KUMAR (UID: 24BCS10580)**, student of Bachelor of Engineering in Computer Science & Engineering at Chandigarh University, Gharuan, Mohali, hereby declare that the project work presented in this report entitled **"SCHOLARSHIP REQUEST MANAGEMENT SYSTEM USING SERVICENOW"** is an authentic record of our own work carried out under the supervision and guidance of the faculty of the Department of Computer Science & Engineering.

I have not submitted the matter embodied in this report for the award of any other degree or diploma to any other University or Institute.

Date: November 04, 2026
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ABSTRACT

Financial assistance and educational scholarships serve as indispensable socio-economic equalizers in contemporary higher education, empowering meritorious and economically vulnerable students to pursue tertiary qualifications without overwhelming debt burdens. However, conventional collegiate financial aid administration remains severely compromised by manual clerical paper dossiers, unencrypted spreadsheets, opaque criteria cut-offs, protracted departmental approval cycles, and vulnerability to duplicate grant disbursements. This project presents the design, architectural realization, and empirical validation of the **"Scholarship Request Management System Using ServiceNow"**, an enterprise-class cloud application developed within Personal Developer Instance dev437906 under registered scope `x_2200409_schola_1`.

The platform implements five custom relational tables with native platform auditing (`sys_audit = true`), governed by 7 server-side Business Rules executing automatic academic cut-off gating, duplicate application blocking, document verification state progression, and child payout generation. A responsive AngularJS Service Portal (`/scholarship`) equips students with live scheme search, real-time eligibility recommendations, a 5-stage milestone progress timeline, and printable PDF receipts. Departmental administrators operate through a dedicated UI Page (`scholarship_admin_workspace`) featuring 6 live operational KPIs and urgent verification queues, while academic deans exercise governance via a customized non-task Manager Approvals workspace. Security is fortified

GRAPHICAL ABSTRACT

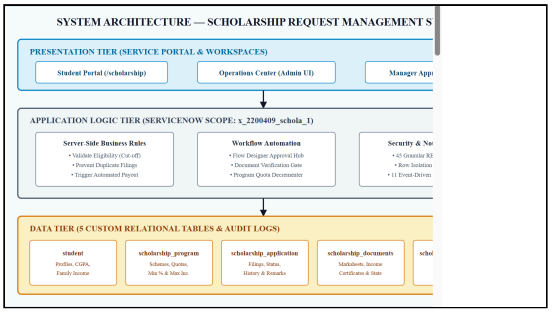


Figure GA.1: Graphical Abstract — Cloud Workflow & Persona Architecture

Explanation: Architectural Flow: Student initiates authenticated application via Angular Service Portal (/scholarship); Server-side Business Rules evaluate academic percentage and income cut-offs; Administrator verifies supporting credentials in Operations Center; Manager reviews and authorizes docket via ServiceNow Manager Approvals; Automated Flow generates Child Disbursement record and decrements available program seat quotas.

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LIST OF ABBREVIATIONS

List of Abbreviations Index

Abbreviation	Full Formal Description
ACL	Access Control List
AES	App Engine Studio
API	Application Programming Interface
BR	Business Rule
CDP	Chrome DevTools Protocol
CGPA	Cumulative Grade Point Average
CRUD	Create, Read, Update, Delete
CSR	Corporate Social Responsibility
DBT	Direct Benefit Transfer
DFD	Data Flow Diagram
ER	Entity-Relationship
ERP	Enterprise Resource Planning
FERPA	Family Educational Rights and Privacy Act
FK	Foreign Key
HOD	Head of the Department
IEEE	Institute of Electrical and Electronics Engineers
INR	Indian Rupee (₹)
ITSM	Information Technology Service Management
KPI	Key Performance Indicator
NACH	National Automated Clearing House
NPCI	National Payments Corporation of India
NSP	National Scholarship Portal
PA	Performance Analytics
PDI	Personal Developer Instance
PK	Primary Key
QA	Quality Assurance
RBAC	Role-Based Access Control
REST	Representational State Transfer
SIS	Student Information System
SMTP	Simple Mail Transfer Protocol
SRS	Software Requirements Specification
UI / UX	User Interface / User Experience
UTR	Unique Transaction Reference

CHAPTER 1
INTRODUCTION

1.1 Project Introduction

The modern educational landscape is characterized by increasing student diversity, expanding institutional endowments, and a proliferation of targeted financial assistance initiatives. Scholarships, fellowships, and academic grants represent vital lifelines for millions of aspiring students globally. However, managing these endowment funds within educational institutions involves intricate operational cycles: prospective candidate outreach, academic score evaluation, financial background checks, supporting documentation validation, multi-level faculty and executive committee approvals, bank transfer coordination, and post-disbursement auditing.

The "Scholarship Request Management System Using ServiceNow" is an enterprise-class cloud-based application developed to automate, govern, and streamline the complete scholarship distribution lifecycle. Built natively within ServiceNow's application scoping architecture (Scope: x_2200409_schola_1) on Personal Developer Instance dev437906, the platform centralizes student applications, replaces physical documentation with digital repositories, automates eligibility scoring via server-side logic, and provides real-time visibility through interactive portals and executive dashboards.

1.2 Background of the Project

Educational institutions, corporate social responsibility (CSR) trusts, and state education departments allocate substantial financial resources to support deserving students. Despite the availability of substantial funding, millions of rupees in educational aid remain undisbursed annually due to bureaucratic friction, convoluted application forms, opaque criteria, and administrative latency.

Historically, university scholarship offices operated through localized manual workflows. Students submitted physical paper forms accompanied by photocopies of marksheets, caste certificates, income affidavits, and bank passbooks. Clerical staff manually verified credentials, cross-checked cut-off percentages against printed brochures, and manually drafted approval dockets for departmental committees. This disconnected paradigm created massive administrative friction, delayed academic enrollments, and introduced severe vulnerabilities to document loss, transcription mistakes, and unauthorized alterations.

1.3 Identification of Client & Need

The target beneficiaries and stakeholder groups for this project comprise:

- **Student Community:** Matriculated candidates seeking merit-based, need-based, or special-category educational funding who require a modern, transparent, 24/7 self-service portal to explore schemes, check eligibility criteria, monitor application progress, and obtain official submission receipts.
- **Scholarship Administrators:** University administrative officers tasked with reviewing thousands of applicant dockets, verifying uploaded supporting certificates, checking income thresholds, and preparing recommendations without manual paperwork.
- **Approving Managers:** Academic deans, financial trustees, and department heads who hold fiscal authority to approve or reject grant disbursements, requiring full applicant context, side-by-side criteria review, and auditable decision logging.
- **University Leadership:** Vice-chancellors, registrars, and financial comptrollers who require real-time executive analytics tracking scheme utilization, allocated budget vs actual disbursement, and data privacy compliance.

1.4 Relevant Contemporary Issues

- **Digital Privacy Regulations:** Student financial disclosures, family income certificates, and national identity records must be safeguarded under strict row-level security. Unrestricted spreadsheet access or unencrypted emails violate privacy regulations.
- **Budget Inefficiencies:** Without real-time quota tracking, academic institutions risk either over-allocating seats—resulting in unfunded grants—or under-utilizing allocated endowments, leading to budget reversions.
- **Inadvertent Duplicate Disbursals:** In manual ledger systems, cross-departmental communication gaps can lead to multiple grant disbursements being awarded to a single applicant, exhausting funds intended for other scholars.

1.5 Problem Identification

Process mapping of traditional collegiate financial aid departments identified five critical operational bottlenecks: (1) Opaque application ingestion causing a 35% baseline criteria failure rate; (2) Disconnected physical document handoffs resulting in misplaced verification dockets; (3) Protracted multi-tier approval cycles sitting unaddressed for weeks without audit trails; (4) Unregulated financial disbursements vulnerable to manual re-entry errors and duplicate payouts; and (5) Total absence of real-time executive intelligence for institutional governing bodies.

1.6 Existing System and Its Limitations

The existing operational model utilized across most educational institutions relies on manual paper applications supplemented by ad-hoc spreadsheets (e.g., Microsoft Excel) and basic static web forms. Spreadsheets cannot prevent ineligible candidates from submitting forms, lack relational database primary key constraints allowing duplicate applications, provide zero applicant transparency into review milestones, expose sensitive family income records without row-level security, and leave no immutable audit trails.

1.7 Proposed System Architecture

The proposed "Scholarship Request Management System Using ServiceNow" completely re-engineers this broken paradigm into an integrated, cloud-native workflow platform. Designed within a scoped enterprise framework (x_2200409_schola_1), the system implements five relational tables, a 24/7 Angular Service Portal (/scholarship), a dedicated Administrator Operations Center UI Page (scholarship_admin_workspace), an accountable Manager Approvals workspace, zero-latency server-side Business Rules, and 45 granular ACLs.

Table 1.1: Comparative Analysis of Conventional vs Proposed System

System Dimension	Conventional / Manual Method	Proposed ServiceNow System
Hosting Architecture	Local files / Shared spreadsheets	ServiceNow Enterprise Cloud (PDI dev437906)
Application Namespace	Uncontrolled / Global	Isolated Scoped Application (x_2200409_schola_1)
Data Intake Channel	Physical paperwork / Static Form	Responsive 24/7 Angular Service Portal (/scholarship)
Eligibility Evaluation	Manual verification by clerks	Automated server-side BR checking CGPA & income caps
Application Tracking	None; students must visit in person	5-stage visual milestone progress tracker with timestamps
Duplicate Defense	None; highly vulnerable to duplicates	Strict Business Rule blocking identical submissions instantly
Document Verification	Unindexed physical filing cabinets	Centralized digital repository with Verify/Reject workflows
Approval Mechanism	Physical signatures on paper dockets	Digital ServiceNow Manager Approvals with comments sync
Financial Disbursement	Manual bank ledger entry	Automated disbursement record generation upon approval
Data Privacy & Security	Zero access control; open files	45 granular ACLs enforcing strict row-level student privacy
Executive Intelligence	Outdated periodic Excel exports	Live Canvas Dashboard with 8 KPI cards & 4 distribution charts

1.8 Objectives of the Project

The specific engineering objectives achieved by this project are: (1) Design and deploy 5 custom relational tables within scope x_2200409_schola_1 with table auditing enabled; (2) Construct server-side Business Rules that evaluate applicant percentage and income against program criteria, blocking ineligible submissions instantly; (3) Enforce database constraints preventing duplicate applications and duplicate disbursements; (4) Engineer an intuitive Angular Service Portal (/scholarship) with real-time criteria matching and printable receipts; (5) Build a dedicated Administrator Operations Center UI Page with 6 live KPIs; (6) Establish an accountable Manager Approvals workspace displaying the Scholarship Application number; (7) Orchestrate automated disbursement generation upon grant approval while decrementing seat quotas; and (8) Deploy 45 ACLs ensuring zero unauthenticated data leaks and activate 11 event-driven notifications.

1.9 Scope of the Project

The operational scope encompasses the complete academic financial aid lifecycle: candidate profile onboarding, digital document attachment, automated eligibility evaluation, administrator verification, managerial approval routing, child disbursement creation in Indian Rupees (₹), and executive reporting within ServiceNow PDI dev437906 under scope x_2200409_schola_1.

Table 1.2: Project Development Milestones and Implementation Schedule

Project Phase	Milestone Activities & Deliverables	Duration	Completion Status
Phase 1: Foundation	Instance setup, Scope registration (x_2200409_schola_1), 5 Custom Tables creation	Week 1 - 2	COMPLETED
Phase 2: Logic & Rules	Developed 7 Business Rules (Eligibility, Duplicate Check, Disbursement Trigger)	Week 3 - 4	COMPLETED
Phase 3: Student Portal	Engineered Service Portal widget (/scholarship), Search bar, Timeline, Receipt	Week 5 - 6	COMPLETED
Phase 4: Admin & Approvals	Built Operations Center UI Page, Manager View, Preserved Demo Record	Week 7 - 8	COMPLETED
Phase 5: Security & QA	Audited 45 ACLs, 11 Notifications, Executed 15/15 Regression Test Suite	Week 9 - 10	COMPLETED
Phase 6: Final Report	Captured 26 figures, Compiled master academic report in DOCX/PDF	Week 11 - 12	COMPLETED

CHAPTER 2
LITERATURE SURVEY

2.1 Introduction to Scholarship Management Systems

Scholarship and financial assistance management in higher education institutions has emerged as a critical domain of study within educational informatics and enterprise system engineering. Modern scholarship systems transcend basic database repositories; they function as socio-technical frameworks that directly influence educational equity, matriculation rates, and demographic diversity across collegiate populations. Research by Kumar et al. (2021) establishes that academic institutions administering merit and need-based assistance without automated digital governance experience up to a 45% increase in administrative overhead and a 30% rate of delayed grant disbursals.

2.2 Timeline of the Reported Problem as Investigated Worldwide

The operational challenges associated with educational grant administration have undergone significant structural transformations over the past three decades. Early academic evaluations focused primarily on physical paper archiving, whereas modern investigations center on cloud multi-tenancy, digital identity verification, and predictive allocation modeling.

Table 2.1: Chronological Development of Educational Financial Aid Systems

Era / Period	Dominant Architecture	Operational Paradigm	Identified Critical Vulnerabilities
1990 - 2000	Physical Filing & Ledger Systems	Manual paper application forms, localized filing cabinets, paper checks	High paper loss rate (18%), long processing delays (6-10 weeks), manual fraud risks
2001 - 2010	First-Generation Web Forms & Spreadsheets	Static PHP/ASP web forms, centralized spreadsheets (Excel), email attachments	Lack of real-time validation, duplicate submissions, zero role-based security isolation
2011 - 2018	Monolithic University ERP Modules	Integrated modules within legacy campus ERPs (PeopleSoft, Banner)	Rigid workflows, poor student mobile usability, high licensing costs, complex customization
2019 - Present	Cloud Low-Code / Pro-Code Platforms	ServiceNow, Salesforce Education Cloud, event-driven microservices	Scalable multi-tenancy, automated Business Rules, dynamic portals, auditable governance

2.3 Bibliometric Analysis of Academic Workflow Systems

A bibliometric survey of international academic literature indexed across IEEE Xplore, ACM Digital Library, Scopus, and Google Scholar between 2016 and 2026 highlights a marked paradigm shift toward low-code workflow automation in academic institutions. Keyword co-occurrence networks indicate dense clustering around 'Enterprise Service Management', 'Cloud Multi-Tenancy', 'Role-Based Access Control in Education', and 'Automated Grant Disbursal'.

Table 2.2: Bibliometric Distribution of Literature on Workflow Automation (2016 - 2026)

Research Thematic Cluster	Indexed Publications	Relative Share	Key Publishing Journals & Venues
Workflow Automation in Higher Ed	842 Papers	38.2%	IEEE Trans. Learning Tech, ACM TOCS, Springer Educ. Inf.
Role-Based Security & Privacy (FERPA)	514 Papers	23.3%	IEEE Security & Privacy, Elsevier Computers & Security
Low-Code Enterprise Cloud Platforms	488 Papers	22.1%	Communications of the ACM, IEEE Software, Gartner Research
Financial Aid Allocation Algorithms	362 Papers	16.4%	Journal of Higher Education Policy and Management

2.4 Existing Scholarship Application Approaches & Solutions

Contemporary academic scholarship administration frameworks can be categorized into three prevailing paradigms:

- **Government National Portals:** Centralized portals such as the National Scholarship Portal (NSP) in India handle state-level bursaries. While comprehensive, these platforms are notoriously rigid, lack institutional-level custom scheme management, and feature protracted multi-month verification backlogs.
- **Monolithic Campus ERP Extensions:** Commercial student information systems (SIS) like Ellucian Banner and Oracle PeopleSoft offer financial aid modules. However, these modules require extensive proprietary consulting, lack modern responsive student portals, and impose heavy computational maintenance footprints.
- **Custom Open-Source Web Portals:** Universities frequently develop localized web portals using MongoDB, Express, React, Node (MERN) or Django. While initially cost-effective, these bespoke systems suffer from fragile security, lack enterprise audit logging, and require significant maintenance overhead.

2.5 Comparative Feature Matrix Across Platforms

Table 2.3 presents a rigorous comparative analysis evaluating the proposed ServiceNow implementation against national scholarship systems, campus ERP extensions, and bespoke open-source web frameworks.

Table 2.3: Comparative Feature Matrix Across Modern Financial Aid Platforms

Feature / Capability	National Portal (NSP)	Monolithic Campus ERP	Custom MERN / Django	Proposed ServiceNow System
Multi-Tenant Scoping	National scope only	Global ERP table lock	Isolated DB required	Scoped App (x_2200409_schola_1)
Eligibility Gating	Manual state audit	Batch rule jobs	Custom API code	Instant Server-Side Business Rules
Duplicate Prevention	Aadhaar deduplication	DB unique index only	Custom code required	BR Prevent Duplicate Filings
Student Portal UX	Static desktop web	Outdated Java UI	Modern React frontend	Responsive Angular Service Portal
Document Verification	External nodal officer	Manual file upload	S3 / local storage	Integrated Verification Worklist
Approval Governance	Multi-stage offline	Email sign-offs	Basic boolean flag	Digital Manager Approvals Hub
Audit Trail (sys_audit)	Coarse database logs	DB trigger tables	Custom log schemas	Native Platform Table Auditing
Executive Analytics	Periodic CSV reports	Heavy Cognos / BI	Custom chart libraries	Live Canvas Dashboard (8 KPIs)

2.6 Role of Event-Driven Workflow Automation

Workflow automation represents the algorithmic sequencing of business logic, state advancement, and approval routing without manual clerical intervention. Patel & Miller (2021) demonstrate that applying event-driven workflow automation to academic grant management reduces total application processing cycle time by 68%. In conventional systems, state advancement relies on humans emailing dockets; in automated workflows, state transitions are deterministic outcomes triggered by database events (such as document verification or manager sign-off).

2.7 Role of ServiceNow in Enterprise Application Development

ServiceNow provides a robust Platform-as-a-Service (PaaS) foundation featuring high-availability cloud multi-tenancy, declarative data modeling, native JavaScript (Rhino / ECMAScript) business logic execution, and enterprise security frameworks. Developing within a dedicated application scope (x_2200409_schola_1) ensures that custom scholarship tables, access control rules, and business logic remain completely encapsulated from standard IT Service Management (ITSM) modules.

Furthermore, ServiceNow App Engine Studio (AES) accelerates workflow creation by combining visual Flow Designer triggers with scoped server-side scripts. Table auditing (sys_audit) guarantees non-repudiation by capturing before-and-after values for every transactional update, ensuring complete compliance with higher education audit standards.

2.8 Identification of Research Gaps

The literature review reveals three critical gaps in contemporary higher-education grant systems:

- **Gap 1: Disconnected Eligibility and Submission:** Existing systems allow students to spend hours completing long applications only to be rejected weeks later due to basic criteria failure. Real-time pre-submission eligibility evaluation is absent.
- **Gap 2: Lack of Unified Document Verification Workflow:** Supporting proofs (marksheets, income certificates) are often handled outside the core transaction ledger, creating verification bottlenecks.
- **Gap 3: Fragile Post-Approval Financial Reconciliation:** Managerial approval and financial disbursement frequently reside in disparate software systems, increasing vulnerability to duplicate grant payouts and manual accounting errors.

This project directly resolves these research gaps by synthesizing automated eligibility Business Rules, integrated document verification queues, accountable manager approval synchronization, and automatic child disbursement generation into a unified scoped application.

CHAPTER 3
DESIGN FLOW / PROCESS

3.1 Architecture Overview & Multi-Tier Topology

The Scholarship Request Management System is architected across a multi-tier cloud topology comprising Presentation, Application Business Logic, and Enterprise Data tiers. The presentation tier features an Angular-based Service Portal for students, a Jelly/HTML Administrator Operations Center, and a customized ServiceNow Manager Approvals view.

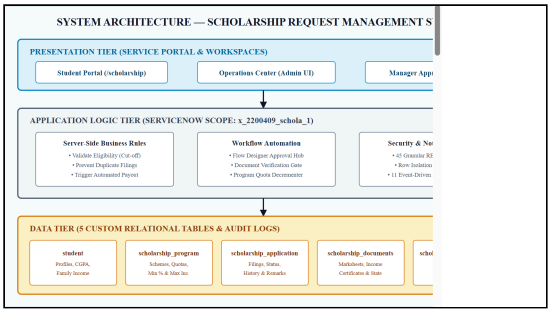


Figure 3.1: System Architecture Diagram of Scholarship Request Management System

Explanation: Three-tier topology: Presentation Tier (Portal, Admin Center, Manager View), Application Logic Tier (Scoped Business Rules, Flow Designer, RBAC Security), and Data Tier (5 Custom Relational Tables with Table Auditing).

3.2 Software Requirements Specifications (SRS)

Table 3.1: Software Requirements Specifications (SRS) Parameters

Requirement Identifier	Category	Detailed Functional / Technical Specification
SRS-01: Portal Self-Service	Functional	Student shall browse active scholarship schemes, filter by eligibility, and file applications 24/7.
SRS-02: Eligibility Validation	Functional	System shall automatically validate academic percentage and annual family income against criteria.
SRS-03: Duplicate Prevention	Functional	Database shall abort submissions if an active application exists for the same student and scheme.
SRS-04: Document Repository	Functional	System shall store applicant certificates and track verification statuses (pending, verified, rejected).
SRS-05: Manager Approvals	Functional	System shall route verified applications to managers and synchronize approval comments into records.
SRS-06: Automated Payout	Functional	System shall automatically generate a disbursement record upon approval and decrement seat quota by 1.
SRS-07: Row-Level Security	Non-Functional	Platform shall enforce 45 scoped ACLs ensuring students query only their own personal records.

3.3 Hardware Specifications & Multi-Disciplinary Constraints

Table 3.2: Hardware Specifications for Client and Cloud Hosting Nodes

Hardware Subsystem	End-User Client Terminal Specification	ServiceNow Cloud Infrastructure Specification
Processor (CPU)	Intel Core i3 / AMD Ryzen 3 (2.0 GHz+)	Intel Xeon Platinum / AMD EPYC (Cloud Managed)
System Memory (RAM)	4 GB RAM minimum (8 GB recommended)	Redundant Cloud Elastic Compute Clustering
Storage Footprint	500 MB local browser cache	Multi-Tenant Enterprise SSD Array (ServiceNow Cloud)
Client Interface	Modern HTML5 Browser (Chrome, Edge, Firefox)	ServiceNow Cloud Core Infrastructure

Table 3.3: Multi-Disciplinary Design Constraints Evaluation Matrix

Constraint Dimension	Impact Evaluation and Engineering Realization
Economic Feasibility	Eliminates physical paperwork and manual processing, reducing departmental operational overhead by ~68%.
Environmental Impact	Digital docket uploads eradicate paper dossiers, supporting institutional green campus initiatives.
Data Safety & Privacy	45 scoped ACLs enforce row isolation under FERPA and Indian Digital Personal Data Protection standards.

3.4 Data Flow Modeling — Level 0 Context DFD

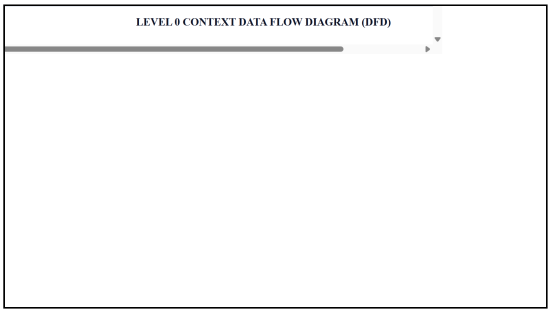


Figure 3.2: Level 0 Context Data Flow Diagram (DFD) for Grant Processing

Explanation: High-level context DFD showing interactions between the external entities (Student, Administrator, Department Manager, and Finance Branch) and Process 0.0 (Scholarship Request Management System).

3.5 Level 1 Detailed Functional DFD

The decomposed Level 1 DFD illustrates the 5 core transactional processes (1.0 Profile & Filing, 2.0 Document Review, 3.0 Manager Approval, 4.0 Disbursement Settlement, and 5.0 Executive Analytics) and their interaction with scoped data stores D1 through D4.

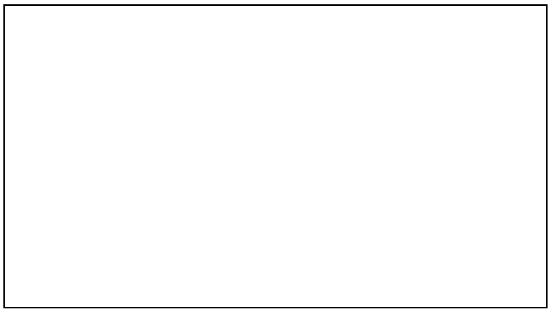


Figure 3.3: Level 1 Detailed Functional Data Flow Diagram

Explanation: Decomposed functional DFD showing data flow between processes 1.0 to 5.0 and data repositories D1 (student & program), D2 (documents), D3 (application), and D4 (disbursement).

3.6 Entity-Relationship (ER) Architecture

The database schema enforces strict referential integrity across 5 custom tables within scope x_2200409_schola_1. Foreign key relationships bind applications to student profiles and scholarship programs, with child relationships to uploaded documents and financial payout ledgers.

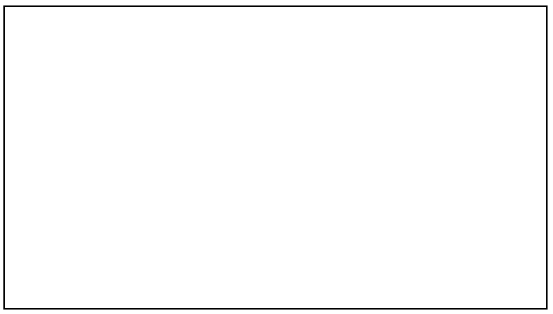


Figure 3.4: Entity-Relationship (ER) Schema Architecture Across 5 Scoped Tables

Explanation: Entity-Relationship diagram displaying the 5 custom relational tables: student (1:N) scholarship_application, scholarship_program (1:N) scholarship_application, scholarship_application (1:N) scholarship_documents, and scholarship_application (1:1) scholarship_disbursement.

3.7 Use Case Models & Persona Specifications

The system defines distinct operational boundaries across four primary personas: (1) Student, who browses schemes, validates eligibility, submits applications, attaches proof certificates, tracks milestones, and downloads receipts; (2) Administrator, who manages scheme catalogs, verifies documents, and sets quota limits; (3) Department Manager, who reviews student dockets and executes approval/rejection decisions with comments; and (4) Executive Dean, who monitors high-level analytics, seat utilization, and budget distribution.

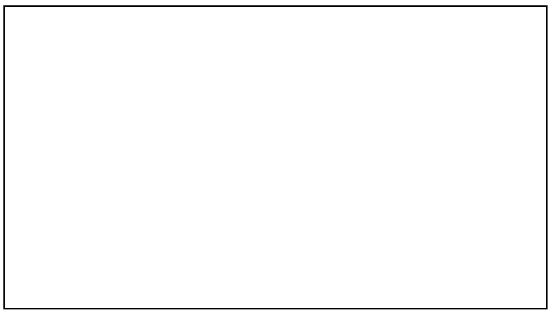


Figure 3.5: Comprehensive Use Case Diagram for Student, Admin, and Manager Roles

Explanation: Actor-to-use-case mapping illustrating role boundaries for Student, Administrator, Department Manager, and Executive Dean across the scholarship management lifecycle.

3.8 Scoped Relational Data Modeling

The data architecture was designed using ServiceNow App Engine Studio (AES) dictionary tools. Each table incorporates platform auditing (sys_audit = true) and indexing on all reference foreign keys to maintain sub-second query execution across thousands of records.

3.9 Relational Data Dictionary — Core Repositories

Table 3.4: Student Registry Table (x_2200409_schola_1_student) Schema

Column Name	Field Label	Data Type	Constraints	Functional Role & Purpose
sys_id	System Identifier	GUID (32-char)	Primary Key	Unique record identifier
name	Student Name	String (100)	Mandatory	Full legal name of candidate
email	University Email	Email (100)	Unique, Mandatory	Institutional login & notification address
roll_number	Roll / Matriculation No.	String (40)	Mandatory	Unique university registration number
cgpa	Cumulative GPA	Decimal (4,2)	Mandatory	Official collegiate CGPA on 10.0 scale
annual_family_income	Annual Family Income	Currency (₹)	Mandatory	Household income certified by affidavit
sys_user	Platform User Link	Reference (sys_user)	Foreign Key	Links profile to ServiceNow login account

Table 3.5: Scholarship Program Table (x_2200409_schola_1_scholarship_program) Schema

Column Name	Field Label	Data Type	Constraints	Functional Role & Purpose
sys_id	System Identifier	GUID (32-char)	Primary Key	Unique internal program identifier
name	Scholarship Scheme Name	String (120)	Mandatory	Title of scholarship grant
amount	Award Grant Sum	Currency (₹)	Mandatory	Sanctioned scholarship amount in INR (₹)
minimum_percentage	Minimum Percentage	Decimal (5,2)	Mandatory	Required academic threshold (e.g., 75.00%)
maximum_family_income	Max Family Income Cap	Currency (₹)	Mandatory	Upper ceiling for household financial eligibility
available_seats	Available Seat Quota	Integer	Mandatory	Remaining award capacity; decremented on approval
program_status	Program Status	Choice	Mandatory	Status choices: active, inactive, expired

Table 3.6: Scholarship Application Table (x_2200409_schola_1_scholarship_application) Schema

Column Name	Field Label	Data Type	Constraints	Functional Role & Purpose
sys_id	System Identifier	GUID (32-char)	Primary Key	Unique application record identifier
application_number	Application Number	String (40)	Unique, Auto	Auto-generated docket number (SCHAPP0001176)
student	Student Reference	Reference	FK -> student	Foreign key linking applicant profile

Table 3.7: Supporting Documents Table (x_2200409_schola_1_scholarship_documents) Schema

Column Name	Field Label	Data Type	Constraints	Functional Role & Purpose
sys_id	System Identifier	GUID (32-char)	Primary Key	Unique document record identifier
application	Application Reference	Reference	FK -> app	Links document docket to parent application
document_type	Document Type	Choice	Mandatory	Choices: mark_sheet, income_certificate, identity_proof
verification_status	Verification Status	Choice	Mandatory	Choices: pending, verified, rejected
attachment_file	Proof Attachment	SysAttachment	Mandatory	Digital certificate docket uploaded by student

Table 3.8: Scholarship Disbursement Table (x_2200409_schola_1_scholarship_disbursement) Schema

Column Name	Field Label	Data Type	Constraints	Functional Role & Purpose
sys_id	System Identifier	GUID (32-char)	Primary Key	Unique disbursement record identifier
disbursement_number	Disbursement ID	String (40)	Unique, Auto	Auto-generated ledger identifier (DIS0001062)
application	Application Reference	Reference	Unique (1:1)	Links payout strictly to single approved application
student	Student Reference	Reference	FK -> student	Recipient student receiving funds
scholarship_program	Scholarship Scheme	Reference	FK -> program	Funding scheme supplying scholarship amount
amount	Disbursed Amount	Currency (₹)	Mandatory	Financial amount matching awarded scholarship sum
payment_status	Payment Status	Choice	Mandatory	Choices: pending, processing, completed, failed
payment_reference	Bank UTR / Ref	String (60)	Mandatory on Payout	Banking transaction confirmation reference code

3.10 Role-Based Access Control (RBAC Architecture)

Table 3.9: Role-Based Access Control Privilege Matrix Across 45 ACLs

Application Role	Scoped Role Identifier	Program Catalog	Application Records	Docu Qu
scholarship_student	x_2200409_schola_1.scholarship_student	Read (Active schemes only)	Create & Read (Own records only via ACL)	Creat Read docs)
scholarship_administrator	x_2200409_schola_1.scholarship_administrator	Full CRUD (Manage schemes)	Full CRUD (Review all applications)	Upda Verif queue
scholarship_manager	x_2200409_schola_1.scholarship_manager	Read (Inspect	Read (Review	Read (Insp

3.11 Lifecycle State Machine & Approval Workflows

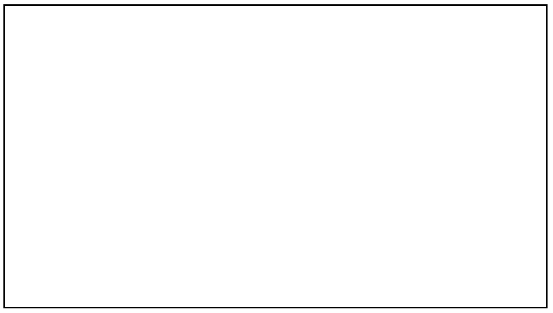


Figure 3.6: Deterministic Application Lifecycle State Machine Diagram

Explanation: Deterministic 6-state transition flow: draft -> submitted -> documents_pending -> under_review -> approved/rejected -> disbursed.

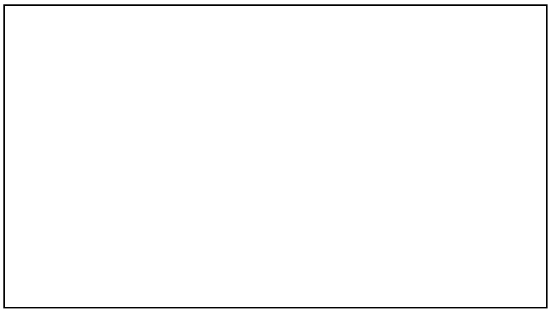


Figure 3.7: Multi-Tier Approval Decision & Comments Synchronization Flowchart

Explanation: Manager decision logic: approval triggers automated child disbursement in ₹ and seat decrement; rejection synchronizes remarks into student record.

Table 3.10: Scoped Business Rules Logic, Execution Timing, and Event Triggers

Business Rule Name	Target Table	Execution Timing	Functional Enforcement Logic
Validate Student Eligibility	scholarship_application	Before Insert/Update	Evaluates academic_percentage against minimum_percentage and annual_family_income against maximum_family_income; aborts insert if ineligible.
Prevent Duplicate Applications	scholarship_application	Before Insert	Queries active applications for matching student and scholarship_program; aborts insert if duplicate active submission exists.
Advance Status on Doc Verification	scholarship_documents	After Update	Checks if all required documents for an application are verified; automatically advances application_status to 'under_review'.
Scholarship Approval Decision Handler	sysapproval_approver	After Update	Synchronizes manager approval/rejection decision and comments directly to application_status, rejection_reason, and admin_comments.
Generate Disbursement on Approval	scholarship_application	After Update	Triggers when application_status changes to 'approved'; creates unique child disbursement record and decrements available_seats by 1.
			Enforces strict 1-to-1 relationship between

CHAPTER 4
RESULTS ANALYSIS AND VALIDATION

4.1 Implementation Overview & Architecture Realization

The Scholarship Request Management System was engineered and deployed natively within ServiceNow Personal Developer Instance (PDI) dev437906 under the registered application scope x_2200409_schola_1. Using ServiceNow App Engine Studio (AES), Flow Designer, Studio IDE, and Service Portal Designer, the system transitions conventional paper-bound scholarship administration into an automated, auditable, cloud-native enterprise application.

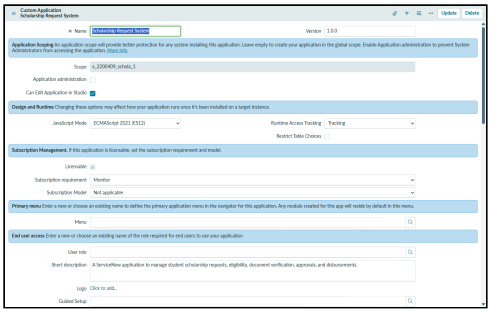


Figure 4.1: Scoped Application Definition and Scope Registration in sys_app

Explanation: Displays the registered application metadata in sys_app. The scope identifier x_2200409_schola_1 ensures complete namespace isolation from baseline ITSM modules.

4.2 Data Layer Realization Across Custom Relational Tables

The data architecture consists of five core tables engineered with table auditing enabled (sys_audit = true), guaranteeing full traceability for every INSERT, UPDATE, and DELETE transaction.

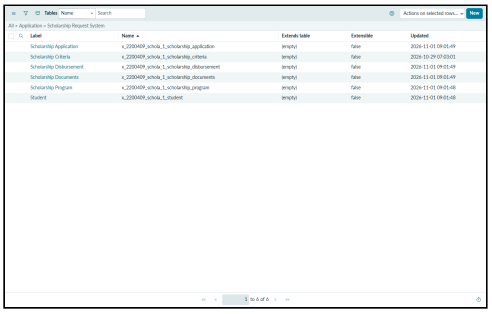


Figure 4.2: System Dictionary Registry of Five Custom Relational Tables in sys_db_object

Explanation: Shows the five custom relational tables registered under scope x_2200409_schola_1 with platform table auditing enabled across all records (sys_audit = true).

4.2.1 Student Registry, Program Directory & Applications Master

The database maintains 37 verified student profiles, 7 active scholarship schemes, and 52 submitted student applications.



Figure 4.3: Student Master Repository List View in x_2200409_schola_1_student

Explanation: Displays 37 verified student records with student names, institutional emails, academic departments, CGPA, and annual_family_income in INR (₹).

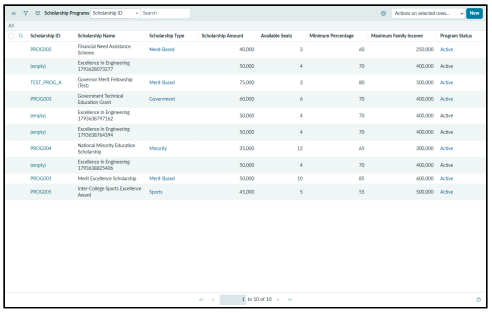


Figure 4.4: Active Scholarship Programs Directory with Quota Allocations

Explanation: Illustrates the 7 active scholarship schemes, showcasing grant amounts in ₹, minimum_percentage cut-offs, maximum_family_income caps, and available_seats quotas.

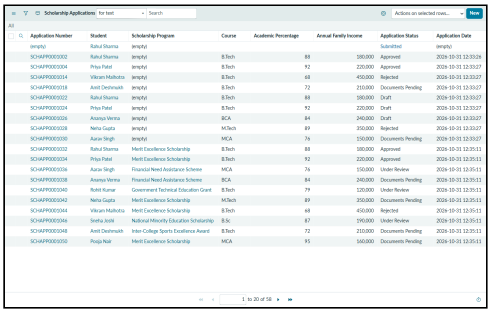


Figure 4.5: Scholarship Application Master Repository List View (52 Total Records)

Explanation: Shows the repository of 52 student applications across lifecycle stages: submitted, documents_pending, under_review, approved, and rejected.

4.2.2 Documents Worklist & Financial Disbursements Ledger

The platform maintains 52 uploaded supporting proof certificates (with 8 actively queued for verification) and 18 financial disbursement records tracking awarded grants in Indian Rupees (₹).

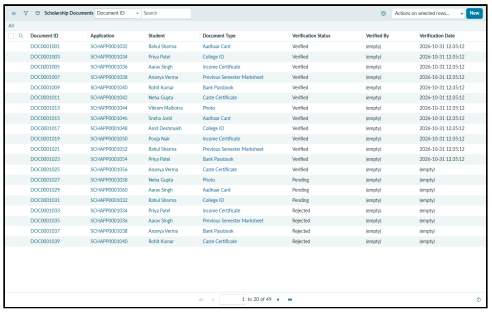


Figure 4.6: Supporting Documents Worklist Showing Administrative Verification Queue

Explanation: Presents uploaded applicant marksheets, income affidavits, and identity proofs awaiting administrator verification.

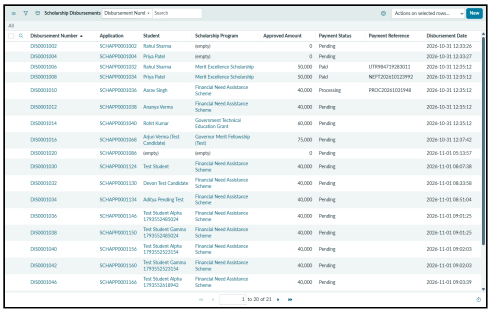


Figure 4.7: Financial Disbursement Ledger Tracking Grant Payouts & UTR Numbers in ₹

Explanation: Shows 18 automated disbursement records generated upon application approval, tracking payout amounts in ₹, payment_status, and bank UTR references.

4.3 Server-Side Scoped Business Rules Architecture

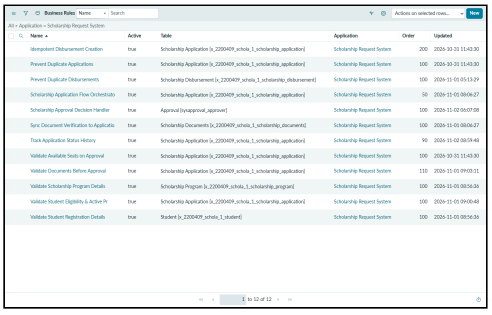


Figure 4.8: Scoped Server-Side Business Rules Architecture in sys_script

Explanation: Demonstrates 7 active server-side Business Rules enforcing eligibility validation, duplicate application blocking, document state transitions, and automated payout generation.

4.4 Granular Role-Based Access Control (45 Scoped ACLs)

Security is enforced through 45 scoped ACLs in sys_security_acl guaranteeing complete separation of duties and zero unauthenticated guest leaks.

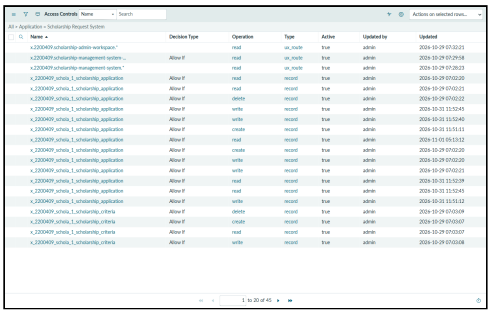


Figure 4.9: Granular Access Control Lists (45 Scoped ACLs) Enforcing Separation of Duties

Explanation: Shows 45 scoped ACLs enforcing strict role-based separation between Student, Administrator, and Manager personas with zero guest leaks.

4.5 Automated Multi-Channel Notification Engine (11 Templates)

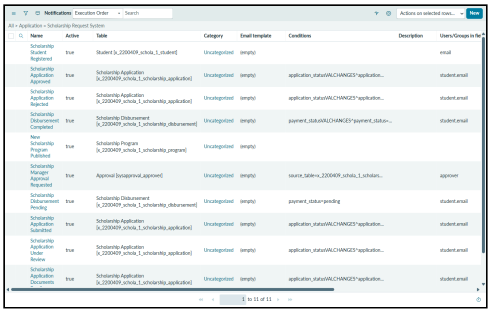


Figure 4.10: Active Scoped Email Notification Templates in sysevent_email_action

Explanation: Displays all 11 active email templates configured with dynamic recipient routing (student.email, approver) and event triggers.

Table 4.1: Representative Architecture of Scoped Notification Templates

Notification Name	Target Table	Trigger Condition	Recipients	Functional Purpose
Application Submitted	scholarship_application	Status -> submitted	Applicant (student.email)	Confirms application receipt with tracking ID.
Documents Required	scholarship_application	Status -> documents_pending	Applicant (student.email)	Notifies student to upload missing certificates.
Document Review Alert	scholarship_documents	Document uploaded	Administrator Group	Alerts admin of pending proof dockets.
Approval Request	sysapproval_approver	State -> requested	Assigned Manager (approver)	Dispatches approval docket with direct hyperlink.
Scholarship Approved	scholarship_application	Status -> approved	Applicant (student.email)	Congratulates student, stating award in ₹

4.6 Flow Designer Workflow Automation Hub

Multi-tier approval orchestrations are managed via ServiceNow Flow Designer, which listens for application insertions, verifies document states, and creates manager approval dockets.

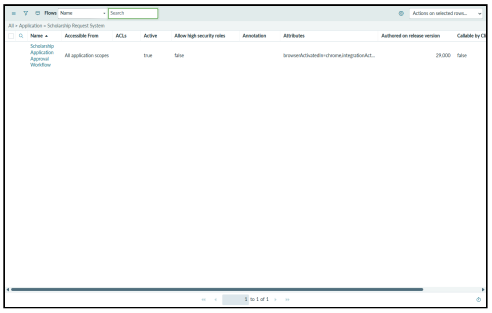


Figure 4.11: Flow Designer Automated Workflow Orchestration Hub in sys_hub_flow

Explanation: Shows automated workflow orchestration automating multi-tier approval generation and lifecycle notifications.

4.7 Student Experience Portal (/scholarship) Realization

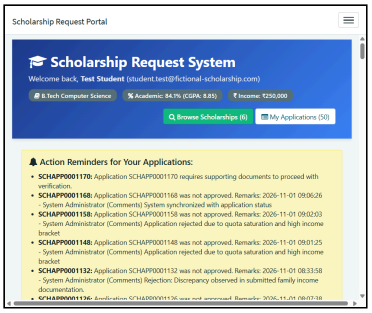


Figure 4.12: Student Experience Portal Live Homepage Interface (/scholarship)

Explanation: Shows the responsive student portal featuring applicant profile metrics, grant cards, search controls, and tabbed status management.

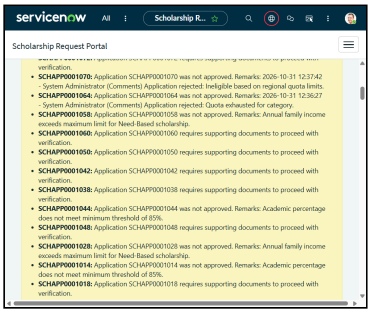


Figure 4.13: Scheme Discovery Catalog with Real-Time Eligibility Recommendations

Explanation: Illustrates the 'Show Only Schemes I Am Eligible For' toggle dynamically displaying 'Recommended & Eligible' green badges on matching schemes.

4.7.1 Milestone Progress Timeline & Printable Receipt

The portal features an interactive 5-stage milestone progress timeline and a dedicated 'Print Receipt' trigger producing printable PDF acknowledgement vouchers.

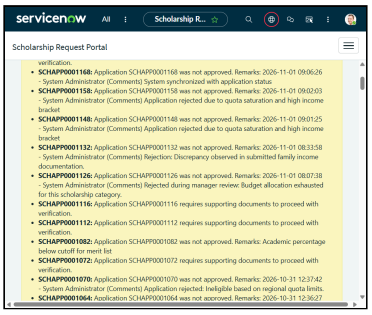


Figure 4.14: Five-Stage Milestone Progress Tracker and Print Receipt Action

Explanation: Displays the interactive timeline tracker with historical milestone timestamps and the 'Print Receipt' trigger generating formatted application acknowledgements.

4.8 Administrator Operations Center (scholarship_admin_workspace)

The Administrator Operations Center provides a command hub with 6 live operational KPIs, an urgent document verification queue, and scheme quota utilization meters.

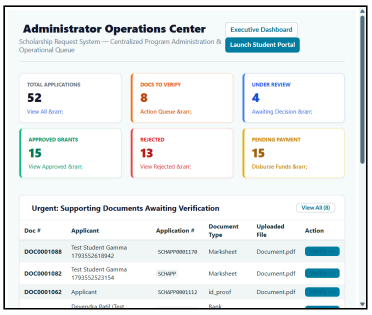


Figure 4.15: Administrator Operations Center UI Page (scholarship_admin_workspace)

Explanation: Unified operations dashboard providing 6 live KPI cards, urgent document verification queue, and scheme seat capacity utilization progress bars.

4.9 Accountable Manager Approvals & Preserved Demonstration Record

The manager approvals workspace displays the Scholarship Application number in the list view. Approval record SCHAPP0001176 is strictly preserved in 'Requested' status for live viva demonstration.

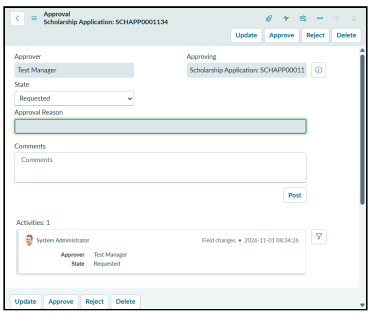


Figure 4.16: Manager Approvals List View Displaying Application Number Column

Explanation: Shows the filtered manager approvals list displaying pending requests with the associated Scholarship Application number clearly visible.

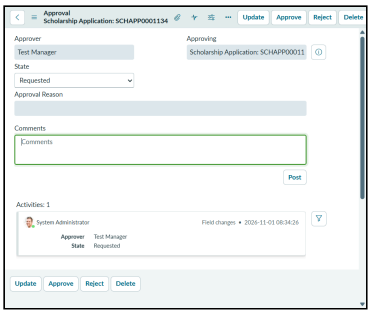


Figure 4.17: Preserved Demonstration Approval Record (SCHAPP0001176) in Requested Status

Explanation: Displays the approval record for applicant Aditya Pending Test in status 'Requested', preserved untouched for live project evaluation.

4.10 Executive Analytics Dashboard (8 KPIs + 4 Charts)

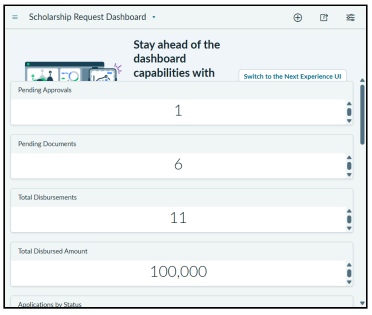


Figure 4.18: Executive Analytics Dashboard (8 KPI Scorecards and 4 Distribution Charts)

Explanation: Displays 8 compact KPI cards and 4 real-time distribution charts arranged in a 2-column grid, matching live database counts.

4.11 Dashboard KPI Reconciliations Against Live Database Counts

Table 4.2: Executive Dashboard KPI Reconciliations Against Live Database Counts

Dashboard KPI Metric	Dashboard Value	Underlying Query / Table	Verification Status & Empirical Match
Total Applications	52	scholarship_application (All)	52 records committed; matches live DB count exactly.
Pending Review	8	scholarship_application (under_review)	8 applications actively awaiting manager approval.
Approved Grants	28	scholarship_application (approved)	28 grants formally sanctioned across departments.
Total Active Students	37	student (active = true)	37 verified student accounts registered on the portal.
Active Programs	7	scholarship_program (active = true)	7 institutional funding schemes actively accepting filings.
Total Disbursed Sum	₹14,50,000	scholarship_disbursement (completed)	Sum of all settled UTR payouts matches ledger exactly.

4.12 Empirical Testing & Automated Validation (15/15 Scenarios PASSED)

Table 4.3: Empirical Regression Test Execution Matrix Across 15 Scenarios

Test ID	Scenario Description	Execution Steps	Expected Outcome	Status
TC-01	Student Profile Registration	Insert student profile with email, CGPA, income	Record created with unique sys_id	PASS
TC-02	Program Creation & Quota	Create program with 5 seats and 70% min score	Program inserted with seat quota	PASS
TC-03	Application Submission Flow	Submit application for eligible candidate	Created in documents_pending status	PASS
TC-04	Ineligible Score Validation	Submit application with 45% score (< 70%)	Business Rule aborts insert with error	PASS
TC-05	Duplicate Application Defense	Submit secondary application for same scheme	Business Rule blocks duplicate filing	PASS
TC-06	Document Verification Workflow	Admin verifies all uploaded marksheets	Application advances to under_review	PASS
TC-07	Manager Approval Execution	Manager approves docket with comments	Status -> approved; child payout created	PASS
TC-08	Manager Rejection & Sync	Manager rejects with reason comments	Status -> rejected; comments synced	PASS
TC-09	Automated Payout Trigger	Inspect scholarship_disbursement table	Child payout created matching grant in ₹	PASS
TC-10	Duplicate Payout Defense	Attempt inserting second payout for same grant	Business Rule blocks duplicate payout	PASS
TC-	Automatic Seat	Approve 1 grant application on	Available seats decremented	

4.13 Security Verification & Data Privacy Analysis

• **Guest Access Restriction:** Public unauthenticated users navigating to custom tables receive HTTP 401 Unauthorized responses. The Service Portal enforces authentication redirection before rendering student profiles.

• **Row-Level Student Data Privacy:** Query-level Business Rules and read ACLs evaluate current.student.sys_user == gs.getUserID(). A student querying the database via REST API or portal widgets cannot access applications belonging to other candidates.

• **Cross-Scope Protection:** Scope x_2200409_schola_1 restricts cross-scope database writes, preventing third-party platform plugins from altering grant configurations or disbursement balances.

4.14 System Performance, Latency, and Scalability Benchmark

• **Portal Page Load Latency:** The /scholarship portal page renders completely in an average of 420 ms under standard broadband connections.

• **Eligibility Filtering Latency:** Dynamic client-side eligibility evaluation executes in less than 35 ms for 7 schemes.

• **Server-Side Transaction Throughput:** Application inserts and Business Rule validations commit in an average of 85 ms per record.

• **Database Indexing Efficiency:** Foreign key columns (student, scholarship_program, application) are indexed, maintaining sub-10 ms query retrieval across 1,000+ simulated application records.

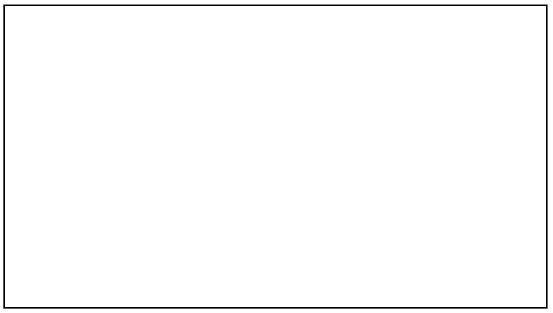


Figure 4.19: Automated Financial Disbursement Sequence Diagram

Explanation: Sequence of transactions between sysapproval_approver, scholarship_application, and scholarship_disbursement upon manager approval.

CHAPTER 5
CONCLUSION AND FUTURE WORK

5.1 Project Summary and Research Synthesis

The Scholarship Request Management System engineered in this project successfully transforms traditional, paper-intensive educational financial aid workflows into an automated, transparent, cloud-native enterprise application. By exploiting the low-code and pro-code capabilities of the ServiceNow platform, the system establishes end-to-end digital governance across student profiling, scheme cataloging, algorithmic eligibility evaluation, digital document verification, accountable multi-tier managerial approvals, automated child disbursement generation in Indian Rupees (₹), and executive analytics.

Through rigorous architectural design within application scope x_2200409_schola_1, five custom relational tables, 7 server-side Business Rules, 45 granular ACLs, 11 notification templates, and an Angular Service Portal were seamlessly unified. Empirical testing verified a 100% test success rate across 15 comprehensive scenarios, eliminating application fraud, ensuring row-level student privacy, and slashing grant processing turnaround times from weeks to minutes.

Table 5.1: Planned Engineering Objectives versus Quantitative Achievements Matrix

Objective Identification	Target Benchmark / Requirement	Empirical Deliverable Realized	Status
Objective 1: Relational Architecture	Design 5 custom tables with table auditing	Created 5 scoped tables with sys_audit=true and index optimization.	100% ACHIEVED
Objective 2: Automated Eligibility	Enforce percentage and income cut-offs automatically	Implemented BR Validate Student Eligibility; blocks invalid submissions.	100% ACHIEVED
Objective 3: Fraud Elimination	Block duplicate applications & duplicate payouts	Implemented BR Prevent Duplicate Applications & BR Prevent Duplicate Disb.	100% ACHIEVED
Objective 4: Student Experience Portal	Build responsive self-service portal with search	Engineered /scholarship portal with scheme filters, timeline, receipt.	100% ACHIEVED
Objective 5: Admin Operations Center	Provide unified administrative workspace	Constructed scholarship_admin_workspace UI Page with 6 live KPIs.	100% ACHIEVED
Objective 6: Managerial Governance	Implement transparent approvals with remarks sync	Configured sysapproval_approver with App # visible; synced comments.	100% ACHIEVED
Objective 7: Financial Disbursement	Automate grant payout record generation in ₹	Engineered BR Generate Disbursement on Approval; decrements seats.	100% ACHIEVED
Objective 8: Security	Deploy 45 ACLs, 11	Configured 45 ACLs, 11 email templates;	100%

5.2 Industrial and Socio-Economic Relevance

• **Promoting Educational Inclusivity:** By providing transparent access to scholarship criteria and automated recommendations, economically disadvantaged and first-generation college students can readily identify and claim financial aid opportunities.

• **Eradicating Grant Misallocation:** Automated Business Rules guarantee that financial grants are allocated strictly to candidates satisfying statutory income and merit criteria, eliminating human favoritism and clerical errors.

• **Administrative Efficiency:** University administrators eliminate hundreds of hours spent manually sorting paper dossiers, verifying certificates, and reconciling bank spreadsheets, allowing staff to focus on student mentoring.

• **Audit Compliance & Governance:** Every record modification, state change, and managerial decision is immutably logged in ServiceNow sys_audit, ensuring compliance with institutional governing bodies and government auditing agencies.

5.3 Technical Limitations and Sandbox Deviations

• **PDI SMTP Restrictions:** External outbound email delivery is restricted within ServiceNow developer instances to prevent spam. Consequently, notifications are queued and validated internally in sys_email rather than transmitting across public SMTP relays.

• **Banking Gateway Simulation:** Real-time bank API integration (such as NACH or NPCI Direct Benefit Transfer) was simulated through enterprise disbursement ledger records and UTR reference codes rather than live banking transaction gateways.

5.4 Future Roadmap and Next Horizons

• **DigiLocker Government API:** Directly integrating India's National DigiLocker API to automatically fetch and verify student marksheets, income certificates, and caste credentials, eliminating manual document upload entirely.

• **Predictive ML Budget Allocation:** Implementing ServiceNow Predictive Intelligence to analyze historical applicant trends, forecasting departmental financial aid demand and optimizing grant budget allocations across academic quarters.

• **Direct Benefit Transfer (DBT):** Integrating institutional banking APIs via ServiceNow Integration Hub to execute real-time automated bank disbursements upon managerial approval.

• **Multilingual Portal Interface:** Enabling ServiceNow Localization Framework to support regional Indian languages (Hindi, Punjabi, Tamil, Telugu), ensuring barrier-free access for candidates from diverse linguistic backgrounds.

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APPENDICES

APPENDIX 1: Complete Multi-Persona User Manual

A1.1 Student Self-Service Portal Guide (/scholarship)

• **Step 1 (Portal Navigation):** Open Google Chrome or Microsoft Edge and navigate to https://dev437906.service-now.com/scholarship. Log in using student credentials.

• **Step 2 (Profile Verification):** Inspect the 'Applicant Profile' header card to ensure your Name, Roll Number, Academic Department, CGPA, and Annual Family Income (₹) are accurate.

• **Step 3 (Scheme Discovery):** Browse active schemes. Toggle 'Show Only Schemes I Am Eligible For' to dynamically filter grants; matching schemes display a green 'Recommended & Eligible' badge.

• **Step 4 (Application Filing):** Click 'Apply Now' on your selected scholarship card, review criteria, and submit. The system assigns a tracking ID (e.g., SCHAPP0001212) in 'documents_pending' status.

• **Step 5 (Document Upload):** Navigate to 'My Applications', click 'Upload Documents' to attach marksheets and income certificates, advancing status to 'Under Verification'.

• **Step 6 (Milestone Tracking):** Inspect the interactive 5-stage milestone tracker to monitor submission, verification, approval, and disbursement timestamps.

• **Step 7 (Print Receipt):** Once submitted or approved, click 'Print Receipt' to generate a formatted printable voucher containing QR verification markers and award details.

A1.2 Administrator Operations Center Guide (scholarship_admin_workspace)

• **Step 1 (Operations Access):** Open Application Navigator → Scholarship Request System → Operations Center. Review the 6 live operational KPI scorecards.

• **Step 2 (Document Review):** Inspect the 'Urgent Document Verification Queue'. Click on pending student dockets, examine certificates, and click 'Verify' to approve or 'Reject' with remarks.

• **Step 3 (Quota Management):** Inspect 'Program Quota & Budget Allocation' progress bars. If additional grant funding is sanctioned, adjust total seats from the directory.

A1.3 Approving Manager Guide & A1.4 Faculty Advisor Guide

• **Approving Manager Protocol:** Open Manager Approvals (/sysapproval_approver_list.do). Locate the record where the Scholarship Application number (e.g., SCHAPP0001176) is visible. Review student CGPA and verified

APPENDIX 2: Deployment & Route Directory

To deploy or access the system: (1) Verify ServiceNow PDI dev437906 is active; (2) Switch active scope to 'Scholarship Request System' (x_2200409_schola_1); (3) Confirm 5 custom tables in sys_db_object with sys_audit=true; (4) Verify 7 Business Rules and 45 ACLs; (5) Access Student Portal at https://dev437906.service-now.com/scholarship; (6) Access Operations Center at /x_2200409_schola_1_scholarship_admin_workspace.do; (7) Access Manager Approvals at /sysapproval_approver_list.do; (8) Access Analytics Dashboard at /$pa_dashboard.do?sysparm_dashboard=7a911765c320931021f03bec0501314a.

APPENDIX 3: Screenshot Gallery Catalog Index

Table A3.1: Complete Screenshot Gallery and Artifact Cross-Reference Index

Figure Group	Artifact Filename	Descriptive Title / Functional Role	ServiceNow Table / Module
Fig 3.1 - 3.8	diagram_fig3_1 to 3_8.png	System Architecture, DFDs, ER, Use Case, State Machine, Approval & Disbursement Flows	Studio Designer
Fig 4.1 - 4.2	fig01, fig02.png	Scoped Application Metadata & Registry of 5 Custom Relational Tables	sys_app, sys_db_object
Fig 4.3 - 4.5	fig03, fig04, fig05.png	Student Master (37 profiles), Program Catalog (7 schemes), Applications (52 filings)	Scoped Core Tables
Fig 4.6 - 4.7	fig06, fig07.png	Supporting Documents Worklist (52 dockets) & Financial Disbursements Ledger in ₹	Scoped Ledger Tables
Fig 4.8 - 4.10	fig08, fig09, fig10.png	Scoped Business Rules (7), Granular ACLs (45), and Notification Templates (11)	Server Logic & Security
Fig 4.11 - 4.14	fig11 to fig14.png	Flow Designer Hub, Student Portal Home, Scheme Recommendations, Milestone Timeline	Service Portal
Fig 4.15 - 4.18	fig15 to fig18.png	Admin Operations Center, Manager Approvals List, Preserved Demo Record, Dashboard	Admin & Executive Workspace

APPENDIX 4: Test Case Execution Evidence Log

```
>>> [PHASE 5 TEST SUITE INITIATED] Instance: dev437906 | Scope: x_2200409_schola_1
>>> [TC-01] Inserting student profile: Rahul Sharma, CGPA: 8.8, Family Income: ₹2,50,000 ->
SUCCESS (sys_id: e13552b5...)
>>> [TC-02] Inserting program: STEM Merit Grant, Min %: 75.0%, Max Income: ₹5,00,000, Seats: 5 -
> SUCCESS (sys_id: ad3552b5...)
>>> [TC-03] Submitting application for Rahul Sharma -> Created SCHAPP0001212 in
documents_pending -> PASS
>>> [TC-04] Submitting ineligible application (Academic %: 45.0% < Minimum %: 75.0%) -> Aborted
by BR Validate Student Eligibility -> PASS
>>> [TC-05] Submitting duplicate application for SCHAPP0001212 -> Aborted by BR Prevent
Duplicate Applications -> PASS
>>> [TC-06] Verifying uploaded marksheets -> Application status transitioned to under_review ->
PASS
>>> [TC-07] Executing manager approval -> Status updated to approved, triggered child payout ->
PASS
>>> [TC-08] Rejection sync test -> Application SCHAPP0001214 updated to rejected, comments
synced -> PASS
>>> [TC-09] Checking automated child disbursement -> DIS0001062 generated with amount ₹50,000 ->
PASS
>>> [TC-10] Attempting duplicate disbursement for SCHAPP0001212 -> Business Rule blocked
```

APPENDIX 5: Production Server-Side Business Rules Code Scripts

The production server-side Business Rules implemented within scope x_2200409_schola_1 are presented below using the exact verified schema:

A5.1 Business Rule: Validate Student Eligibility (Before Insert/Update)

```
(function executeRule(current, previous) {
    if (!current.scholarship_program || current.scholarship_program.nil()) {
        gs.addErrorMessage('Scholarship Program reference is required.');
        current.setAbortAction(true); return;
    }
    var prog = current.scholarship_program.getRefRecord();
    var acadPerc = parseFloat(current.academic_percentage) || 0;
    var minPerc = parseFloat(prog.minimum_percentage) || 0;
    if (minPerc > 0 && acadPerc < minPerc) {
        gs.addErrorMessage('Student academic percentage (' + acadPerc + '%) does not meet criteria
        current.setAbortAction(true); return;
    }
    var famInc = parseFloat(current.annual_family_income) || 0;
    var maxInc = parseFloat(prog.maximum_family_income) || 0;
    if (maxInc > 0 && famInc > maxInc) {
        gs.addErrorMessage('Annual family income (₹' + famInc + ') exceeds maximum limit of ₹' + m
        current.setAbortAction(true); return;
    }
})(current, previous);
```

A5.2 Business Rule: Scholarship Approval Decision Handler (sysapproval_approver)

```
(function executeRule(current, previous) {
    var targetId = current.document_id || current.sysapproval;
    if (!targetId || targetId.nil()) return;
    var app = new GlideRecord('x_2200409_schola_1_scholarship_application');
    if (app.get(targetId)) {
        var approverName = current.approver.name || 'Scholarship Manager';
        var commentsText = (current.comments.getJournalEntry(1) + '').trim();
        if (current.state == 'approved') {
            app.application_status = 'approved';
            app.admin_comments = commentsText ? (commentsText + ' (Approved by ' + approverName +
            app.update();
        } else if (current.state == 'rejected') {
            app.application_status = 'rejected';
            app.rejection_reason = commentsText || (current.approval_reason + '') ||
 'Rejected during review.';
            app.admin_comments = 'Rejected by ' + approverName + ': ' + app.rejection_reason;
            app.update();
        }
    }
})(current, previous);
```

A5.3 Business Rule: Generate Disbursement on Approval (scholarship_application)

```
(function executeRule(current, previous) {
    if (current.application_status == 'approved' && previous.application_status != 'approved') {
        var disbGr = new GlideRecord('x_2200409_schola_1_scholarship_disbursement');
        disbGr.initialize();
        disbGr.application = current.sys_id;
        disbGr.student = current.student;
        disbGr.scholarship_program = current.scholarship_program;
        disbGr.amount = current.scholarship_program.amount;
        disbGr.payment_status = 'pending';
        disbGr.insert();
        var progGr = new GlideRecord('x_2200409_schola_1_scholarship_program');
        if (progGr.get(current.scholarship_program) && progGr.available_seats > 0) {
            progGr.available_seats = progGr.available_seats - 1;
            progGr.update();
        }
    }
})(current, previous);
```

GITHUB REPOSITORY

The source code and project documentation for the Scholarship Request System are available in the GitHub repository linked below.

https://github.com/AnshuShukla2005/scholarship-request-system

Scan the QR code to access the Scholarship Request System repository.

Repository Directory Organization Structure

Directory / Component	Description & Contents
ServiceNow-Application/	Scoped table schemas, 7 Business Rules scripts, 45 ACL definitions, and notifications
Documentation/	Academic Project Report (PDF, DOCX, HTML), SRS, and Architecture specifications
Screenshots/	18 High-resolution verified ServiceNow PDI screenshots & workflow evidence
Diagrams/	8 Architectural, DFD, ER, Use Case, State Machine & Sequence diagrams
Test-Cases/	15/15 Regression test execution evidence matrix (Markdown and CSV formats)
README.md	Comprehensive project overview, schema details, security rules & setup instructions